

C. Inference Using Bayes' Rule

- i) From the definition of Conditional Probabilities and Commutativity of Conjunction we can derive the following Product rules:

$$P(a \wedge b) = P(a \mid b) * P(b), \text{ where } P(b) > 0, \text{ and}$$

$$P(a \wedge b) = P(b \mid a) * P(a), \text{ where } P(a) > 0.$$

- ii) From the product rules above we get the **Bayes' rule/ law/ theorem**,

$$P(b \mid a) = P(a \mid b) * P(b) / P(a), P(a) > 0 \text{ and } P(b) > 0,$$

which **underlies all modern AI systems for probabilistic inference.**

[Probability of hypothesis given the evidence/ effect]

Example 1.

Say, the following facts are known.

'toothache' is caused by 'cavity' 60% of times : $P(\text{toothache} \mid \text{cavity}) = 0.6$

1 in 10 patients investigated has 'cavity' : $P(\text{cavity}) = 1/10 = 0.1$

1 in 4 patients investigated has 'toothache' : $P(\text{toothache}) = 1/4 = 0.25$

We can now derive: $P(\text{cavity} \mid \text{toothache}) =$

Example 2.

Given,

$$P(\text{Water} \mid \text{Coal}) = 0.4,$$

$$P(\text{Coal}) = 0.2,$$

$$P(\text{Water}) = 0.8.$$

So, $P(\text{Coal} \mid \text{Water}) =$

D. Inference using Joint Probabilities of Consistent Models of the Environment

Recall the Monster that Smells:

? 1,3			
OK S 1,2	? 2,2		
OK 1,1	OK S 2,1	? 3,1	

- ✓ 3 reachable squares[fringe/ frontier]: [1,3], [2,2], [3,1]; 3 random variables: $M_{1,3}$, $M_{2,2}$, $M_{3,1}$;
- ✓ Pure logical inference can help no more;
- ✓ Probabilistic agent can do much better after computing $P(M_{1,3})$, $P(M_{2,2})$, $P(M_{3,1})$;
- ✓ We assume that OK means 'No Monster', and independent probability of a Monster at any unknown cell is 0.2.

Consistent models of the Environment for fringe variables $M_{1,3}$, $M_{2,2}$, $M_{3,1}$:

.....

$$\text{i) } P(M_{1,3} \mid \text{evidence}) = \langle P(m_{1,3} \mid \text{evidence}), P(\neg m_{1,3} \mid \text{evidence}) \rangle$$

$$\text{ii) } P(m_{1,3} \mid \text{evidence}) =$$

$$\frac{\text{Sum of the joint probabilities of models where } m_{1,3} \text{ holds}}{\text{Sum of the joint probabilities of all consistent models}} \approx 0.31$$

$$\text{iii) } P(\neg m_{1,3} \mid \text{evidence}) \approx 0.69$$

$$\text{iv) } P(M_{1,3} \mid \text{evidence}) \approx \langle 0.31, 0.69 \rangle = \langle 31\%, 69\% \rangle$$

$$P(M_{2,2} \mid \text{evidence}) \approx \langle 0.86, 0.14 \rangle = \langle 86\%, 14\% \rangle \quad [\text{Self study}]$$

$$P(M_{3,1} \mid \text{evidence}) \approx \langle 0.31, 0.69 \rangle = \langle 31\%, 69\% \rangle \quad [\text{Self study}]$$

So, which cell is safer to move to?